

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

23

Q1

ALGEBRA

A rocket contained 467,000 kilograms (kg) of propellant before launch. Exactly 21 seconds after launch, 362,105 kg of this propellant remained. On average, approximately how much propellant, in kg, did the rocket burn each second after launch?

- A. 4,995
- B. 17,243
- C. 39,481
- D. 104,895

In science class, Diego conducted an experiment to learn about evaporation. Diego measured the height of fluid in a beaker over a period of time. The function $f(x) = 39 - 0.18x$ gives the estimated height, in centimeters cm, of the fluid in the beaker x days after the start of the experiment. Which of the following is the best interpretation of 39 in this context?

- A. The estimated height, in cm, of the fluid at the start of the experiment
- B. The estimated height, in cm, of the fluid at the end of the experiment
- C. The estimated change in the height, in cm, of the fluid each day
- D. The estimated number of days for all the fluid to evaporate

x	y
1	5
2	7
3	9
4	11

The table above shows some pairs of x values and y values. Which of the following equations could represent the relationship between x and y ?

A. $y = 2x + 3$

B. $y = 3x - 2$

C. $y = 4x - 1$

D. $y = 5x$

Q4

ALGEBRA

For what value of w does $w - 10 = 2(w + 5)$?

- A. 5
- B. 0
- C. -15
- D. -20

Q5

ALGEBRA

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 7 and passes through the point $(0, 5)$. Which equation defines f ?

- A. $f(x) = 5x$
- B. $f(x) = 35x$
- C. $f(x) = 7x + 5$
- D. $f(x) = 12x + 5$

$$4x + 3y = 24$$

Mario purchased 4 binders that cost x dollars each and 3 notebooks that cost y dollars each. If the given equation represents this situation, which of the following is the best interpretation of 24 in this context?

- A. The total cost, in dollars, for all binders purchased
- B. The total cost, in dollars, for all notebooks purchased
- C. The total cost, in dollars, for all binders and notebooks purchased
- D. The difference in the total cost, in dollars, between the number of binders and notebooks purchased

GRID-IN ALGEBRA

What value of x is the solution to the given equation?

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

ALGEBRA — SET 23 · ALGEBRA

Area (square feet)	Water (gallons)
2,520	4,536
3,780	6,804
5,040	9,072

The buildings of a shopping center are designed to allow water to drain from the roof into gutters on the sides of the buildings. The table shows the relationship between the area x , in square feet, of a roof and the amount of water fx , in gallons, drained from the roof into the gutters over a certain period of time. Which equation could define f ?

- A. $fx = 0.6x$
- B. $fx = 1.8x$
- C. $fx = 2,268x$
- D. $fx = 4,536x$

In a chess tournament, each participant earns 1 point for each game the participant plays that ends in a draw and 3 points for each game the participant wins. A certain participant in this tournament has earned 41 points. Which equation represents this situation, where d represents the number of games this participant has played that ended in a draw and w represents the number of games this participant has won?

A. $d + 3w = 41$

B. $3d + w = 41$

C. $d + \frac{w}{3} = 41$

D. $\frac{d}{3} + w = 41$

Q10

GRID-IN

ALGEBRA

If $6n = 12$, what is the value of $n + 4$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.